

Personality Type Predictor: Executive Summary

What the project delivers

A local English Streamlit app predicts one of four dataset personality-type labels from 19 questionnaire responses and three demographic inputs. The project includes EDA, leakage-aware preprocessing, model tuning, MLflow tracking, and an explicit handoff of the selected trained pipeline. It is a demonstration, not a diagnosis or a complete standard Big-Five test.

Main findings and decisions

Cleaning. The raw data contain 19,719 records. Removing three complete duplicates and one all-zero questionnaire record leaves 19,715. The 83 age values above 100 become missing; missing demographic values are retained for training-pipeline imputation. Retaining a record does not establish the validity of all its remaining responses.

Class balance. Moderate accounts for 42.86% of cleaned records and Resilient for 31.26%; Overcontroller and Undercontroller account for 14.35% and 11.54%. Stratification preserves proportions. Balanced class weights change training error weights; macro-F1 gives each class equal evaluation weight. These are distinct measures, not interchangeable forms of balancing.

Model comparison. Three model families receive 20 Hyperopt trials each with stratified five-fold training cross-validation. Their best mean CV macro-F1 scores are Random Forest 0.786, Logistic Regression 0.784, and KNN 0.744. Logistic Regression is selected for explainability and its competitive CV results, including slightly higher training out-of-fold Undercontroller F1. The small mean difference does not prove statistical superiority. KNN tuning was a later completeness follow-up; the known holdout was not used to reopen the champion decision.

Final evaluation. The fixed champion achieves accuracy 0.824 and macro-F1 0.784 on 3,943 holdout records. Undercontroller precision is 0.455, recall 0.699, and F1 0.551. Thus fewer than half of predictions of this label are correct, although approximately 70% of its actual members are recognized. Resilient F1 is 1.000 in this dataset, not evidence of perfect generalization. The holdout is used to report limitations, not to tune or select a new model.

Application. Inputs remain raw until the fitted pipeline applies its stored imputation, scaling, and encoding. The app never trains a new model. Ages outside 13-100 are rejected before prediction. Sliders, clear labels, explicit errors, and a technical input view support understandable operation. Neutral defaults are demonstration inputs, not a validated average profile.

Reproduction and evidence

Datasets, models, MLflow stores, and virtual environments are not published. Follow the README for data access, setup, notebook execution, and the new run-ID handoff. EDA and modeling hold the saved evidence; the project report explains the decisions in more detail.

MLflow evidence. The saved champion run is 7d27288ce33e41d3b5490f8b04e9b07d (Logistic Regression). It links the fitted pipeline and training-CV metrics; the notebook reports holdout macro-F1 0.784. This is historical evidence, not a supplied model. Reproduction creates a new ID to verify and hand to the app.

Next step. Evaluate transferability on independent data with compatible inputs and documented labels. This is proposed future work, not a completed test.

For demonstration only, not for clinical or other consequential decisions.

Source repository: github.com/der-berg/personality-type-predictor